

Study Plan — Patient-Level Skin-Cancer Risk Modelling from Nevus Phenotypes + Automated Photodamage

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Purpose: combine nevus-related phenotypes with the automated whole-body photodamage score (`severity_score` , 0–2) from the DermLIP photodamage model into a patient-level skin-cancer risk model. This document sets out the objective, risk-group definitions, predictors, statistical approach, and sample-size requirements.

1. Objective & hypotheses

Primary question: Does an objective, automated whole-body photodamage score add predictive value for skin-cancer risk *beyond* nevus phenotypes and standard demographics?

- **H1 (association):** higher `severity_score` is associated with higher-risk status.
- **H2 (incremental value):** adding photodamage to a nevus + demographics model improves discrimination ($\Delta\text{AUC} > 0$).
- **H3 (interaction, exploratory):** photodamage and naevus burden act synergistically — e.g. the high-photodamage × high-naevus subgroup is disproportionately high risk.

2. Cohorts, data sources, and design

2.1 Available cohorts

Cohort	Description	Used to train the sun-damage model?	Role in the risk study
HOP (sun-damage development dataset)	from the BJD paper; ~100 patients, predominantly the HOP study. Only a subset of the full HOP cohort was used for the sun-damage model.	Yes (main training source)	risk-model training (see leakage note below)
MYM (high-risk melanoma cohort)	to be added for the risk study; enriched for skin-cancer events	partly / to confirm	risk-model training + testing (split)
185 VIC (AUTOMATE-ME)	new Victorian-sites cohort; no overlap with HOP or MYM	No	risk-model training + testing (split); clean out-of-sample photodamage scores

2.2 Data strategy for the risk model

The sun-damage development dataset alone (~100 patients) is **too small** to develop a robust multivariable risk model, so we pool cohorts:

- **Risk-model training** = sun-damage development dataset (HOP) + a portion of MYM and/or 185 VIC.

- **Risk-model testing / validation** = **held-out** patients (remaining MYM / 185 VIC) and/or **newly collected cohorts** — patient-disjoint from the training set.

2.3 severity_score provenance — leakage note (important)

The photodamage `severity_score` is a *predicted* feature, so where it is predicted matters:

- **HOP patients were in the sun-damage model's training set** → their `severity_score` is an **in-sample** prediction (mildly optimistic). Acceptable for **risk-model training**, but to be rigorous, generate **out-of-fold** severity predictions (cross-fit the sun-damage model) for these patients, or keep HOP for training only.
- **MYM / 185 VIC were not seen by the sun-damage model** → their `severity_score` is a genuine **out-of-sample** prediction. **Reserve these (or new cohorts) for the risk-model test set** so the reported risk-model performance uses honest photodamage inputs. (This is what the plan does.)

2.4 Phasing

- **Phase 1 — pilot / development:** pool HOP + partial MYM/185 → develop the risk model; internal validation.
- **Phase 2 — external validation:** held-out MYM/185 and/or newly collected cohorts.

3. Defining the risk groups

Three options, recommended in this order of preference.

Approach	High risk	Low risk	Notes
A. Outcome-anchored (preferred)	history of melanoma or ≥1 keratinocyte carcinoma (BCC/SCC), or incident skin cancer during follow-up	no skin-cancer history / incidence	Needs outcome / pathology data across the pooled cohorts — the gold standard for a <i>risk</i> model. Applied within cohorts, not across, to avoid confounding.
B. Cohort-defined risk (known-groups proxy)	MYM high-risk melanoma cohort	population / VIC cohort	MYM is enriched for risk by design, so this is a ready label — but it is confounded by cohort/site (batch effects), so treat as a supportive discrimination check, not the primary endpoint.
C. Composite-score tertiles (exploratory)	top tertile of a derived risk index	bottom tertile	Data-driven; use to describe, not to claim prediction.

Recommendation: use **A** as the **primary endpoint** if skin-cancer outcome/history data exist; use **B** only as a supportive known-groups check (with the cohort-confounding caveat); keep **C** exploratory. Dichotomise only for reporting — model the underlying continuous risk rather than binarising predictors.

⚠ Key input required: what outcome data exists across the pooled cohorts (prior melanoma/KC flags, incident lesions, follow-up time). If outcomes are scarce, Phase 1 falls back to association + the option-B known-groups check. If option B is used, adjust for cohort/site to avoid attributing site batch-effects to photodamage.

4. Predictors — how the phenotypes enter the model

Sun-damage model outputs - `severity_score` (0–2, continuous) — primary photodamage variable - per-body-region severity (anterior / posterior torso, upper / lower limbs) — site-specific damage - fraction of severe tiles; pigmentation score (secondary)

Nevus phenotypes - total nevus count (log-transform — counts are right-skewed) - atypical / dysplastic nevus count; count of nevi > 5 mm; nevus density / distribution

Standard covariates (confounders) - age, sex, Fitzpatrick skin type, hair / eye colour, freckling, personal / family melanoma history

Handling: standardise continuous predictors; check collinearity (photodamage correlates with age — keep both but monitor VIF); allow non-linearity via restricted cubic splines for age and nevus count; pre-specify the photodamage × nevus interaction for H3.

5. Statistical modelling plan

1. **Primary model:** multivariable **logistic regression** (outcome = high vs low risk) with pre-specified predictors. Report odds ratios (95% CI), AUC, calibration.
 2. **Incremental value (key analysis for H2):** hierarchical nested models — `demographics` → + `nevus phenotypes` → + `photodamage` — compared with **DeLong Δ AUC**, **likelihood-ratio test**, and **NRI / IDI**. Directly answers "does photodamage add anything beyond nevi?"
 3. **If predictors are many relative to events:** use **penalised regression (elastic net / LASSO)** rather than stepwise selection.
 4. **Longitudinal (Phase 2, if incident outcomes):** Cox proportional-hazards, time to first lesion.
 5. **Internal validation:** bootstrap (≥ 500 reps) for optimism-corrected AUC and calibration slope; report per **TRIPOD**.
 6. **Output:** a risk score / nomogram; then define risk tertiles for clinical communication.
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6. Sample size

The driver is **events (cancers / high-risk cases)**, not total patients.

Why the sun-damage development dataset alone is not enough

The sun-damage dev dataset covers only **~100 patients**. For a multivariable model the rule is $\geq 10\text{--}15$ **events per predictor (EPV)**: with ~10 candidate predictors you need **100–150 events**. At an assumed skin-cancer prevalence of ~30–40 %, ~100 patients → ~30–40 events → **EPV $\approx 3\text{--}4$** → **underpowered / overfitting**. Hence the plan to **pool HOP + MYM + 185 VIC** for training.

Phase 1 — pooled development set

- **Association / group comparison (H1):** to detect a moderate difference in `severity_score` between high- and low-risk groups (Cohen's $d \approx 0.4\text{--}0.5$) at 80 % power, $\alpha = 0.05$ → **~65–100 patients per group**. The pooled set (HOP ~100 + a portion of MYM/185) comfortably supports this for a **pilot**; adequacy for a *multivariable* model depends on total **events**, not patients (see below).

- **Pilot multivariable model:** keep the predictor set small (3–5) and use **penalised regression** until the pooled event count reaches $EPV \geq 10$.

Phase 2 — target for a robust model

- Applying **Riley et al. (2020)** minimum-sample-size logic (~10 predictors, prevalence ~0.3, target C-statistic ~0.75) → roughly **400–600 patients with ≥ 120 –150 events** for stable development, plus a **separate external validation set of ≥ 100 events** (~200+ patients). This is the concrete rationale for pooling HOP + MYM + 185 VIC for training and **holding out remaining MYM/185 or new cohorts** for testing.

Exact numbers can be computed (`pmsampsize`) once three inputs are fixed: **(a) expected outcome prevalence, (b) final predictor count, (c) target C-statistic.**

7. Phased roadmap

Phase	Data	Deliverable
1 (now, $n \approx 185$)	current cohort + nevus counts + severity scores	pilot: association, MYM-vs-HOPS discrimination, incremental-value signal (penalised), power/feasibility read
2 ($n \approx 500$ –800, multi-site)	pooled sites + linked outcomes / follow-up	full risk model, TRIPOD-reported, internal + external validation, risk-group thresholds

8. Key limitations to state up front

- Photodamage correlates with age — must adjust, or the model simply re-learns age.
- Cross-sectional Phase 1 shows association, not causation / prediction of *incident* disease.
- Cohort is surveillance-enriched (MYM) — risk estimates need recalibration for general populations.

9. Open inputs needed before finalising

1. **Outcome data** across the pooled cohorts (melanoma / KC history, incident lesions, follow-up time) — for HOP, MYM, and 185 VIC.
2. **MYM cohort** — how many patients, and do they have nevus counts + the imaging needed to generate `severity_score` ?
3. Which patients go to **risk-model training vs held-out testing** (confirm the split and that HOP is used for training only, or generate out-of-fold severity for HOP).
4. **Nevus phenotype fields** available and their definitions (total count, atypical, size bins).
5. Sample-size assumptions: **outcome prevalence, predictor count, target C-statistic.**

References

- Riley RD, *et al.* Calculating the sample size required for developing a clinical prediction model. *BMJ* 2020;368:m441.
- Collins GS, *et al.* TRIPOD statement: transparent reporting of a multivariable prediction model. *Ann Intern Med* 2015;162:55–63.
- Pencina MJ, *et al.* Evaluating added predictive ability of a new marker (NRI / IDI). *Stat Med* 2008;27:157–72.
- Peduzzi P, *et al.* A simulation study of the number of events per variable in logistic regression. *J Clin Epidemiol* 1996;49:1373–9.

Companion to `V4_RESULTS_REPORT.pdf` (model results) and the deployment package `model_package/photodamage-dermlip-mtl-v4/` (`severity_score` definition).